Project Proposal
CS4100 Artificial Intelligence (Fall 2025) with Professor Rajagopal Venkat
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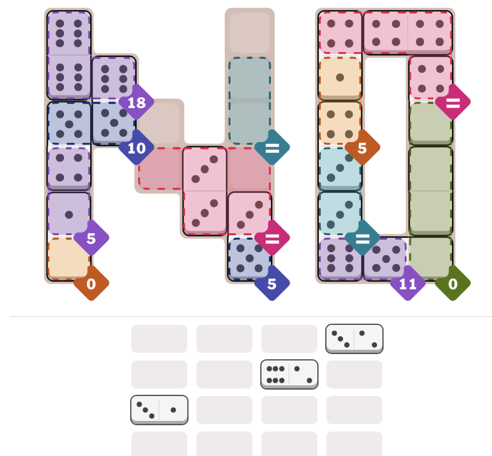
Cracking the Code: First AI System for NYTimes Pips



NYTimes Pips, launched in August 2025, is a single-player domino puzzle requiring players to arrange tiles on a grid while satisfying multiple regional constraints. Each colored region specifies requirements: sum constraints (pips must total a specific value), equality constraints (all pips must match), or inequality constraints (individual pips must be greater/less than a threshold). Unlike traditional domino games focused on matching, Pips demands simultaneous satisfaction of overlapping mathematical constraints. Pips offers an unexplored computational challenge with clear difficulty scaling: Easy (4-5 dominoes), Medium (8-10 dominoes), and Hard (14-16 dominoes).

As daily Pips players, we’ve experienced the challenge and satisfaction of cracking Hard puzzles and the “aha” moments when constraint patterns suddenly click. This personal engagement drives our interest in uncovering the game’s underlying complexity. Beyond our enthusiasm, Pips represents a perfect testbed for fundamental AI techniques with real-world applications. The constraint satisfaction skills transfer directly to circuit design (component placement), workforce scheduling (shift assignments with requirements), and logistics optimization (cargo loading with restrictions). By developing an AI solver, we bridge the gap between a game we love and practical algorithmic challenges, potentially discovering strategies that neither human intuition nor pure computation would find alone. Successfully automating Pips would not only enhance our own gameplay but demonstrate how AI can augment human problem-solving in constraint-heavy domains.

While no published work exists specifically for Pips, related puzzle domains offer valuable insights. CSP algorithms like Arc Consistency (AC-3) successfully solve Sudoku but must be adapted for Pips' overlapping regions. Recent advances in Monte Carlo Tree Search for puzzle games show promise but haven't addressed domino placement under multiple constraint types. One unverified YouTube video claims to demonstrate an AI solving Pips, but provides no technical details or reproducible methodology. Our approach will be the first documented, open-source solution.

We will investigate three AI techniques from our coursework to determine the optimal solving strategy for Pips: Constraint Satisfaction Problems (CSP), local search algorithms, and reinforcement learning. CSP naturally models Pips' structure with domino placements as variables and regional requirements as constraints. Local search techniques like simulated annealing will address computational bottlenecks by iteratively minimizing constraint violations through domino repositioning. Reinforcement learning, specifically Q-learning, will explore whether agents can learn effective placement strategies through self-play.